Ambient WiFi as a Radar Replacement for Automotive SLAM: A Physics-Based Feasibility Study

Mulham Fetna

***Abstract*—Automotive radar is a costly, dedicated sensor, while WiFi transceivers are ubiquitous and, under the integrated sensing and communication (ISAC) paradigm, increasingly dual-purpose. We ask whether *ambient* sub-7 GHz WiFi—received on a single moving-vehicle antenna and processed with commodity channel-state information (CSI)—can serve in place of radar as the perception front-end of a SLAM pipeline. Using physics-based ray tracing (Sionna RT), we build an end-to-end pipeline: ray-traced multipath channel $\rightarrow$ super-resolution (MUSIC) delay/angle estimation $\rightarrow$ bistatic reflector triangulation $\rightarrow$ Rao–Blackwellized particle-filter SLAM, and evaluate it with trajectory (ATE/RPE) and map (Chamfer, occupancy IoU, directional accuracy/completeness) metrics. We find that ambient WiFi *localizes* the vehicle to centimetre level with a clean operating envelope that follows the $\Delta R = c/2B$ resolution law, and that a joint two-dimensional (delay–angle) MUSIC front-end lifts *realistic* commodity-CSI localization to oracle-sensing quality in sparse-multipath scenes. In contrast, *mapping* separates into two tiers: with clean single-bounce (oracle) sensing the back-end reconstructs reflecting surfaces to 25–30 cm, whereas realistic sensing is bounded to 4–5 m. Counter-intuitively, this floor is removed by *neither* a $44\times$ bandwidth increase (60 GHz) *nor* a $4\times$ larger array: the limit is not resolution but *path discrimination*—telling a genuine reflection from a line-of-sight or floor path—which commodity CSI does not provide. We show this discrimination is *learnable in principle* from per-path physical features (F1 ≈ 0.9 when an arrival-elevation feature is available; note that a single-ULA delay–azimuth front-end does not estimate elevation, so this is an upper bound rather than a commodity-CSI result), delineating precisely what passive WiFi can and cannot do for automotive SLAM. To support further work we release the full pipeline and WiFiSLAM-Sim, the first ray-traced outdoor/vehicular WiFi-CSI dataset with per-path labels.**

***Index Terms*—Internet of Things, WiFi sensing, passive WiFi radar, channel state information, SLAM, integrated sensing and communication (ISAC), connected and autonomous vehicles, automotive perception, multipath-based SLAM.**

I. Introduction

AUTOMOTIVE perception relies heavily on radar for its robustness to weather and lighting, but radar is a dedicated, cost- and power-bearing sensor. In parallel, the built environment is saturated with WiFi: access points, roadside units and connected vehicles form a dense, ambient IoT connectivity fabric. Under the integrated sensing and communication (ISAC) paradigm and the IEEE 802.11bf sensing amendment, this same communication infrastructure is increasingly dual-purpose—turning already-deployed radios into opportunistic sensors at no extra hardware cost [1], [2], [3]. For connected and autonomous vehicles, exploiting this ambient IoT fabric for perception is especially attractive. This raises a concrete question: *can ambient WiFi, received on a moving vehicle, replace radar as the perception input to SLAM?*

The gap. A two-round, adversarially fact-checked survey (summarized in Sec. II) exposes a clean and defensible gap. Passive WiFi radar and CSI-based sensing are mature but almost exclusively *indoor, static and infrastructure-fixed* [4], [5], [6], [7]; the one demonstrated *moving-platform* passive radar uses a 5G downlink—not WiFi—as its illuminator [8]; and joint communication and SLAM is explicitly "in its infancy" [9]. No prior work occupies the intersection of an *on-vehicle, mobile, outdoor, WiFi-band* sensing system feeding a SLAM pipeline.

The physical fork. Range resolution is bandwidth-limited, $\Delta R = c/2B$: commodity sub-7 GHz WiFi resolves only 3.75 m at 40 MHz and 0.94 m at 160 MHz, whereas radar-grade resolution ($\sim$8.5 cm) requires 60 GHz mmWave (IEEE 802.11ad DMG) [10], [1]. A WiFi-for-SLAM design must therefore either (i) stay sub-7 GHz and extract coarse geometry from multipath, or (ii) move to 60 GHz for fine range. *This paper takes the sub-7 GHz path and characterizes it honestly and end-to-end*, isolating what is achievable now from what needs the mmWave extension.

Contributions.

- An open, reproducible, physics-based (Sionna RT ray-traced) end-to-end pipeline for on-vehicle ambient-WiFi SLAM (Fig. 1), spanning channel simulation, super-resolution sensing, bistatic mapping and particle-filter SLAM.
- A demonstration that ambient sub-7 GHz WiFi *localizes* a moving vehicle to centimetre level, with an operating envelope (bandwidth, SNR, access points, speed) that quantitatively follows the $\Delta R = c/2B$ law.
- A *joint two-dimensional (delay–angle) MUSIC* front-end whose intrinsic association lifts realistic commodity-CSI localization to oracle-sensing quality in sparse multipath.
- A two-tier characterization of *mapping*: an oracle upper bound (25–30 cm) and a realistic bound (4–5 m), with the residual gap shown—by a 60 GHz and a 16-antenna test—to be a *path- discrimination* limit of commodity CSI rather than a bandwidth or aperture (resolution) limit, identifying the key open problem for WiFi mapping.

M. Fetna is an independent researcher (robotics/mechatronics). E-mail: contact@mulhamfetna.com. ORCID: 0009-0006-4432-798X.

Code, configurations and results are openly available and citable: GitHub https://github.com/mulhamfetna/wifi-radar-slam; Zenodo concept DOI 10.5281/zenodo.21247288.

- **WiFiSLAM-Sim**, the first (ray-traced) outdoor/vehicular WiFi-CSI dataset with per-path labels, and a learned discriminator that predicts mapping-useful reflections from per-path features (F1 ≈ 0.9 at realistic noise, using an arrival-elevation feature that a single-ULA front-end does not measure; it is therefore an upper bound)—a first concrete step on the path-discrimination problem.

II. Related Work

A. Passive WiFi Radar

Ambient WiFi access points act as illuminators of opportunity for bistatic passive radar. Chetty *et al.* first demonstrated through-wall detection of moving personnel, extracting both range and Doppler [4]. Because WiFi bursts have short integration time and low bandwidth, standard Fourier processing is inadequate, motivating super-resolution (ESPRIT/MUSIC) methods [5]; passive radar has been shown against unmodified stand-alone access points [6], and phase-robust CSI-power processing achieves sub-metre indoor tracking [11]. All of this work is indoor and static—the point of departure for a mobile outdoor system.

B. WiFi CSI Sensing and 3D Reconstruction

WiFi-band signals reflect off bodies and penetrate walls; learning-based methods reconstruct dense structure—2D/3D human pose [12], [13], [7], [14], depth [15], and even latent-diffusion imagery [16]. These systems prove dense WiFi "imaging" is possible, but every one is a few-metre *fixed indoor* zone with poor cross-environment generalization [7]—which both motivates and bounds the present work. We are careful to distinguish purpose-built WiFi-band FMCW radios [12] from decoded commodity 802.11 CSI [14].

C. RF- and Multipath-based SLAM

The SLAM back-end we require already exists. Channel-SLAM treats each multipath component as a line-of-sight signal from a *virtual transmitter*, jointly estimating landmark and receiver states with no prior map [17], [18]; modern multipath-based SLAM extends this with cooperation, map fusion and belief propagation, including real mmWave-MIMO measurements [19], [20]. We reuse this virtual-transmitter / bistatic-reflector landmark model and test its robustness to a fast-moving receiver.

D. ISAC, IEEE 802.11bf and mmWave WiFi

ISAC formalizes the shared-hardware sensing/communication tradeoff [2], [21], [3], and IEEE 802.11bf standardizes the sensing measurement plumbing while leaving algorithms to vendors [1]. Crucially for our design fork, the 802.11ad 60 GHz waveform has been reused as a radar with <0.1 m range resolution and V2I imaging, albeit in theory/simulation [10], [22]. This frames our sub-7 GHz-now, 60 GHz-next roadmap.

III. System Model and Method

A. Scenario and Bistatic Geometry

A vehicle carrying a single receive antenna array drives through an outdoor scene illuminated by N_{AP} ambient WiFi access points at known positions. Each specular multipath component follows an access point $\rightarrow$ reflector $\rightarrow$ vehicle path, so the measured path length $\rho = \|\mathbf{p}_{\mathrm{AP}} - \mathbf{r}\| + \|\mathbf{r} - \mathbf{p}_{\mathrm{veh}}\|$ places the reflector $\mathbf{r}$ on an ellipse with foci at the access point and the vehicle. Given the arrival azimuth ϕ (the bearing from the vehicle to $\mathbf{r}$), the reflector range s along $\hat{\mathbf{u}}(\phi)$ solves

$$s = \frac{\rho^2 - d_{\mathrm{AP}}^2}{2\big(\rho - \mathbf{d}_{\mathrm{AP}} \cdot \hat{\mathbf{u}}(\phi)\big)}, \qquad \mathbf{d}_{\mathrm{AP}} = \mathbf{p}_{\mathrm{AP}} - \mathbf{p}_{\mathrm{veh}}, \quad (1)$$

with $d_{\mathrm{AP}} = \|\mathbf{d}_{\mathrm{AP}}\|$; the direct path ($s \leq 0$) and physically implausible solves are rejected.

B. Ray-Traced Channel Model

Channels are generated with Sionna RT 2.0 (Mitsuba/Dr.Jit), which ray-traces the scene (buildings, vehicles) to produce per-path delays, angles, interaction types and the frequency-domain channel (CFR) across N_{sc} OFDM subcarriers and N_{rx} antennas, per vehicle pose along the trajectory. Additive white Gaussian noise is applied at a configured SNR.

C. Sensing Front-End

We compare two sensing tiers. The *oracle* front-end reads Sionna's true single-scatter path parameters (delay, azimuth), isolating the mapping/SLAM geometry from estimation error. The *realistic* front-end estimates parameters from the noisy CSI with MUSIC. For delays we use the antennas as snapshots over a physically-bounded delay grid; for azimuth we use the subcarriers as snapshots. A key refinement is *joint 2-D (delay–angle) MUSIC* with 2-D spatial smoothing (SVD subspace; delay grid bounded to the unambiguous $1/\Delta f$ range), which recovers each path's delay *and* angle together, so their association is intrinsic rather than paired by sorted index. Array-relative electrical angles are mapped to world azimuth by $\sin\beta = -\sin\theta$ for the vehicle's lateral ULA.

D. SLAM Back-End and Metrics

A Rao–Blackwellized particle filter propagates the vehicle pose from odometry and weights particles by the bistatic consistency (1) of each detection; triangulated reflectors are accumulated and consensus-filtered into a landmark map. We report trajectory error (ATE, RPE) and, for the map, symmetric Chamfer distance, occupancy IoU, and *directional* accuracy (estimate $\rightarrow$ ground truth) and completeness (ground truth $\rightarrow$ estimate), since a passive-WiFi map illuminates only a subset of surfaces. Ground truth for the map is the facade footprint of each scene object.

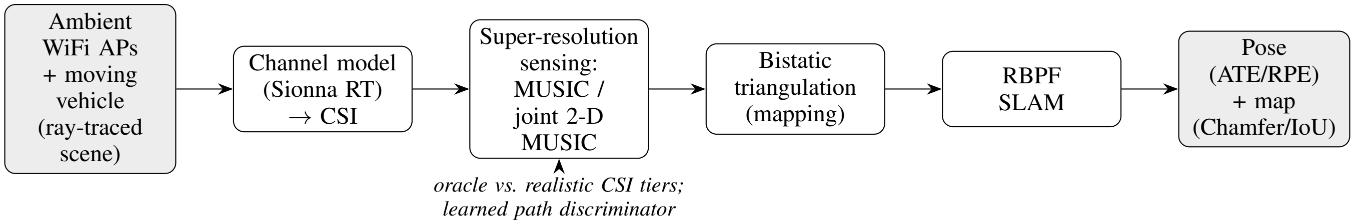

Fig. 1. End-to-end ambient-WiFi SLAM pipeline. Ambient access points illuminate a moving vehicle in a ray-traced scene; the channel-state information is processed by a super-resolution sensing front-end (MUSIC, or joint 2-D delay–angle MUSIC; evaluated under an oracle and a realistic tier, with an optional learned path discriminator) into bistatic reflector detections, which a Rao–Blackwellized particle filter fuses into a vehicle pose and a reflector map.

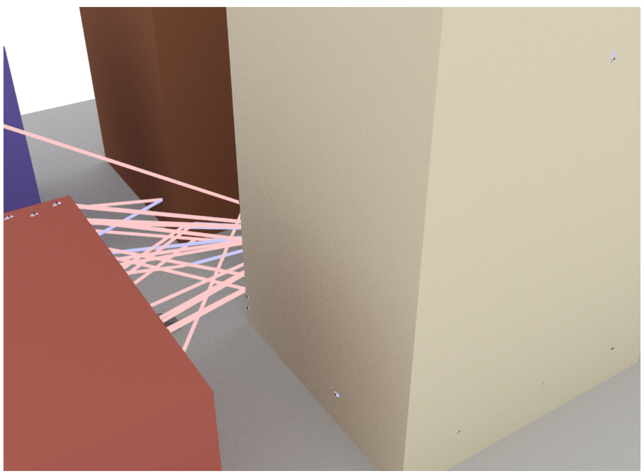

Fig. 2. Ray-traced street-canyon scene with the ambient-WiFi multipath rays from an access point to the vehicle antenna (Sionna RT render).

TABLE I
LOCALIZATION OPERATING ENVELOPE: ABSOLUTE TRAJECTORY ERROR (ATE, M), ONE PARAMETER SWEPT AT A TIME ABOUT THE NOMINAL POINT.

Bandwidth		SNR	
20 MHz	0.78	0 dB	2.02
40 MHz	0.093	10 dB	1.86
80 MHz	0.058	20 dB	1.01
160 MHz	0.038	30 dB	0.076
# Access points		Speed	
1	0.084	1 m/s	2.13
2	1.97	5 m/s	0.032
3	0.075	10 m/s	0.047
4	0.041	15 m/s	0.010

IV. Experimental Setup

Two scene families are used. A *street canyon* (Sionna's `simple_street_canyon_with_cars`) provides a realistic, dense-multipath outdoor environment; a *controlled reflective* scene (a scaled metal wall) provides a geometry in which clean single-bounce returns are guaranteed, isolating sensing from illumination effects. Unless stated otherwise: carrier 5.2 GHz, 40 MHz bandwidth, $N_{\text{sc}} = 64$, $N_{\text{rx}} = 4$ (half-wavelength ULA), 3 access points, 20 dB SNR, 5 m/s straight trajectory. Simulations run CPU-only (Sionna RT LLVM backend) on a 32-thread workstation; each Phase-A run completes in tens of seconds. All numbers below are reproduced from the released result artifacts (`docs/results-v1.md`).

V. Results

A. Localization Operating Envelope

Ambient WiFi localizes the vehicle to centimetre level. Table I sweeps the four principal parameters. Bandwidth shows the expected monotonic ∼20× ATE improvement from 20 MHz to 160 MHz, tracking the $\Delta R = c/2B$ law end-to-end; SNR degrades gracefully to sub-decimetre by 30 dB; more access points and higher speeds are generally beneficial.

B. Realistic Sensing: Joint Delay–Angle MUSIC

Separate 1-D delay and 1-D angle estimates paired by sorted index mis-associate in multipath and scatter phantom reflectors. Joint 2-D MUSIC removes this by estimating each path's delay and angle together. In the controlled scene at 160 MHz, realistic ATE improves from 0.73 m (sorted) to 0.098 m ± 0.028 m (joint; mean ± std over five seeds), a ≈ 7× gain that *approaches but does not match* the 0.049 m ± 0.027 m oracle bound. Because the pipeline is stochastic, we report a seed spread rather than a single run. The benefit is scene-dependent: in the dense street canyon (∼19 paths, 4 antennas) the paths cannot be resolved and joint estimation does not help, so it is retained as an opt-in front-end.

C. Mapping: Oracle vs. Realistic

Table II contrasts the two sensing tiers. With oracle sensing the map reconstructs reflecting surfaces to 25–30 cm (Fig. 3, left); the same geometry under realistic MUSIC sensing is bounded to 4–5 m. Critically, this residual persists even in the front/back-unambiguous controlled scene and even under joint estimation; as Sec. VII shows, it is removed by neither wider bandwidth nor larger aperture. The bound is therefore a *path-discrimination* limit—genuine facade reflections cannot be separated from the consistent LOS/floor paths commodity CSI returns without an interaction label—not a resolution problem.

D. Real Commodity-CSI Proof-of-Concept

To verify that the sensing front-end is not simulation-specific, we ran the same MUSIC delay/AoA estimator on measured hardware CSI through a CSIKit-backed adapter. On Intel 5300 captures (30 subcarriers, 3 antennas) and Broadcom

TABLE II
MAPPING ACCURACY: ORACLE VS. REALISTIC SENSING.

Scene / sensing	accuracy (m)	Chamfer (m)	IoU
Controlled wall, oracle	0.25	0.51	0.79
Street canyon, oracle	0.30	12.3 †	0.08
Controlled wall, realistic (joint)	4.8	4.1	0.0

† symmetric Chamfer inflated by partial coverage; a single WiFi pass illuminates only street-facing facades.

TABLE III
NEITHER BANDWIDTH NOR APERTURE MOVES REALISTIC MAP ACCURACY.

Configuration (controlled wall, joint)	accuracy (m)	ATE (m)
sub-7 GHz, 160 MHz, 4-antenna	4.8	0.098
60 GHz, 1.76 GHz, 4-antenna	4.8	0.030
60 GHz, 1.76 GHz, 16-antenna	4.75	0.034

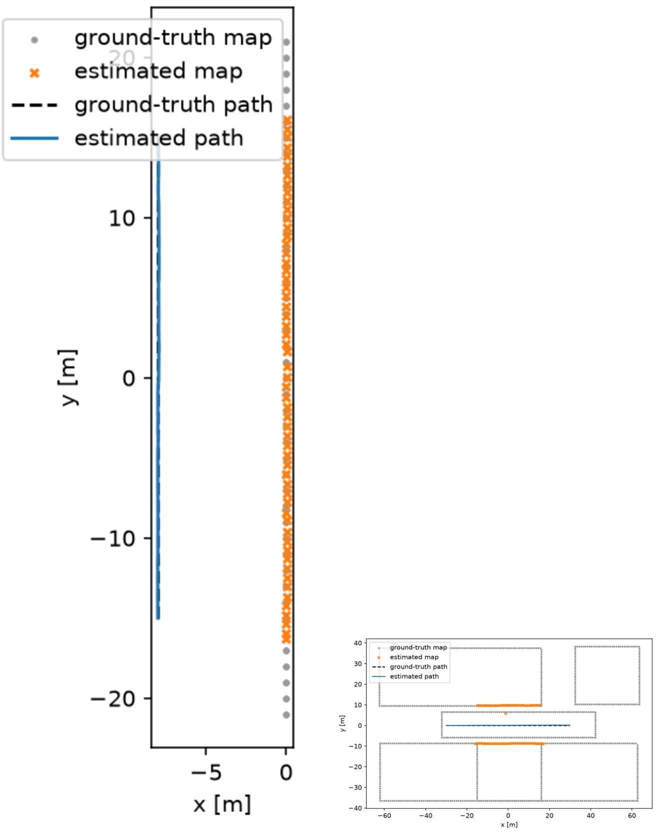

Fig. 3. Oracle maps. Left: controlled wall—estimated reflectors (orange) trace the wall to ∼0.25 m. Right: reflective street canyon—reflectors trace the two illuminated street-facing facades; distant/outer surfaces are uncovered (coverage, not accuracy, limited).

nexmon 802.11ac captures (80 MHz), the front-end runs end-to-end and returns plausible indoor multipath delay/angle estimates. This is a front-end validation only—the captures are indoor and static with no ground-truth trajectory—but it confirms the pipeline consumes real commodity CSI. A full outdoor, vehicle-mounted validation is blocked by the absence of any public outdoor/vehicular WiFi-CSI dataset (the very gap this work targets) and is the key next step.

VI. DISCUSSION

The results delineate a clear boundary. *Localization* from ambient sub-7 GHz WiFi is practical today: centimetre-level under oracle sensing and decimetre-level from realistic commodity CSI with joint estimation. *Mapping* is fundamentally harder, but—as Sec. VII shows—not for the reason one would expect: the realistic floor is removed by neither wider bandwidth nor larger aperture, so it is not a resolution limit but a *path-discrimination* limit. Genuine facade reflections cannot be separated from the consistent line-of-sight and floor paths that commodity CSI returns without an interaction label. For deployment, the practical implication is encouraging: a connected vehicle can obtain a cm-level pose from the ambient WiFi it already receives, with no dedicated sensor and no new spectrum—an attractive low-cost complement to (or graceful fallback for) radar in the IoT-connected fleet—while environment mapping awaits the discrimination advances above. Limitations of this study are its simulation-primary scope (mitigated by the real-CSI front-end check of Sec. V) and the single-array receiver.

VII. DOES MORE BANDWIDTH OR APERTURE FIX MAPPING? A 60 GHZ TEST

The intuitive remedy for the mapping floor is finer resolution: a 60 GHz (IEEE 802.11ad DMG) front-end offers ∼1.76 GHz of bandwidth ($\Delta R \approx 8.5\,cm$, a $44\times$ improvement), and a larger array sharpens angle. We tested both directly by re-running the controlled-wall pipeline at 60 GHz and with a 16-element array (Table III).

Strikingly, neither change moves the map accuracy (Fig. 4). The reason is visible in the reconstructed map: the estimated reflectors form a *consistent phantom arc along the vehicle trajectory*, not the wall—these are the line-of-sight and floor-bounce paths, which MUSIC returns among the strongest components at every pose and which, being geometrically consistent, survive consensus filtering. Bandwidth and aperture sharpen *resolution*, but the mapping floor is *path discrimination*: distinguishing a genuine facade reflection from an LOS/floor/multi-bounce path. The oracle succeeds only because it can filter by the ray tracer's interaction type; commodity CSI carries no such label.

Consequence. The realistic-mapping limit is neither the $\Delta R = c/2B$ resolution ceiling nor array aperture—so 60 GHz alone does not enable passive-WiFi mapping. It is the absence of bounce/path discrimination in commodity CSI, which we identify as the key open problem. Localization, which does not require this discrimination, is unaffected and remains centimetre-level at both bands.

As a first step toward discrimination, we gate on the bistatic *excess* path length (LOS and floor paths barely detour, so their excess is near zero). This halves the phantom error (map accuracy 4.8 m→2.0 m) but collapses coverage, because the delay bias makes genuine reflections low-excess too and they are gated out alongside the phantoms.

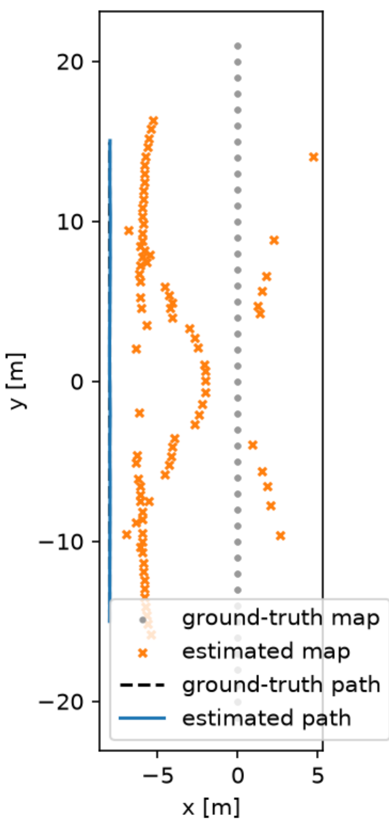

Fig. 4. Realistic 60 GHz map (joint MUSIC). Despite 8.5 cm range resolution, the estimated reflectors (orange) form a consistent phantom arc along the vehicle trajectory (LOS/floor paths), not the wall (grey, at $x{=}0$).

A. Learned Path Discrimination

Is the discrimination *learnable*? We release **WiFiSLAM-Sim**, a ray-traced outdoor/vehicular WiFi-CSI dataset that—uniquely, being simulated—carries a per-path oracle label (bounce count, interaction type, floor flag) usable as a supervised target. A random forest predicts the binary label *"mapping-useful single-scatter facade reflection"* from per-path features a receiver can estimate (range, bistatic excess, azimuth, elevation, azimuth-deviation-from-AP)—not the interaction type. On exact features the classes are perfectly separable (F1 $=$ 1.00, excess and azimuth-deviation dominating); under realistic MUSIC-level feature error it degrades gracefully—F1 $=$ 0.94/0.90/0.86 at range/azimuth noise of $(1\,\text{m}, 3°)/(2\,\text{m}, 6°)/(4\,\text{m}, 10°)$—well above the fixed excess gate. Path discrimination is thus *learnable and robust*; its ceiling is set by feature-estimation accuracy (chiefly the delay-derived excess), closing the loop with the delay-bias finding.

VIII. Future Work

Three directions follow directly. *(i) Feature estimation for path discrimination*: our learned discriminator is accurate given estimable features, so the remaining problem is estimating them (chiefly the delay-derived excess) robustly from noisy CSI—e.g. end-to-end learning directly on CSI rather than on MUSIC outputs. *(ii) A real outdoor, vehicle-mounted WiFi-CSI capture*: WiFiSLAM-Sim supplies a simulated dataset with labels, but real-hardware outdoor CSI still does not exist and is the prerequisite for full-system (not just front-end) validation. *(iii)* Tighter integration with a virtual-transmitter multipath-SLAM back-end [17], [19] and Doppler exploitation for moving clutter.

IX. Conclusion

We presented an open, physics-based feasibility study of ambient sub-7 GHz WiFi as a radar replacement for automotive SLAM. Ambient WiFi localizes a moving vehicle to centimetre level—and to oracle-sensing quality from realistic CSI via joint delay–angle MUSIC in sparse multipath—while mapping separates into an oracle upper bound (25–30 cm) and a realistic bound (4–5 m). We further showed that this mapping floor is removed by neither a 60 GHz wide-band front-end nor a larger array—it is a *path-discrimination* limit of commodity CSI, not a resolution one. The study thus maps the boundary of the achievable for passive WiFi perception and identifies bounce/path discrimination as the central open problem for WiFi-based mapping.

Reproducibility and Data Availability

The complete pipeline (channel simulation, MUSIC/joint-MUSIC sensing, bistatic mapping, particle-filter SLAM, and the learned discriminator), all scene/run configurations, and every result artifact are openly available under AGPL-3.0: GitHub https://github.com/mulhamfetna/wifi-radar-slam, versioned and archived on Zenodo (concept DOI `10.5281/zenodo.21247288`). Each table/figure is reproduced by a single command from a released config. The **WiFiSLAM-Sim** dataset is generated by `make_dataset.py` and documented by a datasheet; a sample is attached to the release.

References

[1] "An overview on IEEE 802.11bf: WLAN sensing," *IEEE Communications Surveys & Tutorials*, 2024. [Online]. Available: https://arxiv.org/abs/2310.17661

[2] F. Liu, Y. Cui, C. Masouros, J. Xu, T. X. Han, Y. C. Eldar, and S. Buzzi, "Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 6, pp. 1728–1767, 2022.

[3] Zhang *et al.*, "Joint communication and sensing (JCAS): a survey," arXiv preprint arXiv:2102.12780, 2021. [Online]. Available: https://arxiv.org/abs/2102.12780

[4] K. Chetty, G. E. Smith, and K. Woodbridge, "Through-the-wall sensing of personnel using passive bistatic WiFi radar at standoff distances," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 50, no. 4, pp. 1218–1226, 2012. [Online]. Available: https://ieeexplore.ieee.org/document/6020778

[5] A. Yildirim, H. Griffiths *et al.*, "Super-resolution passive radars based on 802.11ax Wi-Fi signals for human movement detection," *IET Radar, Sonar & Navigation*, vol. 15, no. 4, pp. 323–339, 2021. [Online]. Available: https://ietresearch.onlinelibrary.wiley.com/doi/full/10.1049/rsn2.12038

[6] W. Li, R. J. Piechocki, K. Woodbridge, C. Tang, and K. Chetty, "Passive WiFi radar for human sensing using a stand-alone access point," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 3, pp. 1986–1998, 2021. [Online]. Available: https://discovery.ucl.ac.uk/id/eprint/10103371

[7] Y. Yan *et al.*, "Person-in-WiFi 3D: End-to-end multi-person 3D pose estimation with Wi-Fi," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2024/papers/Yan_Person-in-WiFi_3D_End-to-End_Multi-Person_3D_Pose_Estimation_with_Wi-Fi_CVPR_2024_paper.pdf

[8] R. Maksymiuk *et al.*, "5G-based passive radar on a moving platform—detection and imaging," *IET Radar, Sonar & Navigation*, 2024. [Online]. Available: https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rsn2.12559

[9] Gounis, S. A. Tegos, D. Tyrovolas, P. D. Diamantoulakis, and G. K. Karagiannidis, "When SLAM meets wireless communications: A survey," arXiv preprint arXiv:2602.06995, 2026. [Online]. Available: https://arxiv.org/abs/2602.06995

[10] P. Kumari, J. Choi, N. González-Prelcic, and R. W. Heath, "IEEE 802.11ad-based radar: An approach to joint vehicular communication-radar system," *IEEE Transactions on Vehicular Technology*, vol. 67, no. 4, pp. 3012–3027, 2018. [Online]. Available: https://arxiv.org/abs/1702.05833

[11] Wang, Zhang, Wu, Chen, Xu, and Guo, "Bistatic passive sensing via CSI power," arXiv preprint arXiv:2511.22144, 2025. [Online]. Available: https://arxiv.org/abs/2511.22144

[12] M. Zhao, T. Li, M. Abu Alsheikh, Y. Tian, H. Zhao, A. Torralba, and D. Katabi, "Through-wall human pose estimation using radio signals," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. [Online]. Available: https://rfpose.csail.mit.edu/

[13] M. Zhao, Y. Tian, H. Zhao, M. Abu Alsheikh, T. Li, R. Hristov, Z. Kabelac, D. Katabi, and A. Torralba, "RF-based 3D skeletons," in *Proceedings of the ACM SIGCOMM Conference*, 2018.

[14] F. Wang, S. Zhou, S. Panev, J. Han, and D. Huang, "Person-in-WiFi: Fine-grained person perception using WiFi," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019. [Online]. Available: https://www.ri.cmu.edu/app/uploads/2019/09/Person_in_WiFi_ICCV2019.pdf

[15] "Wi-depth: Depth reconstruction of moving objects from WiFi CSI," arXiv preprint arXiv:2503.06458, 2025. [Online]. Available: https://arxiv.org/abs/2503.06458

[16] "LatentCSI: CSI-to-image synthesis via a pretrained latent diffusion model," arXiv preprint arXiv:2506.10605, 2025. [Online]. Available: https://arxiv.org/abs/2506.10605

[17] C. Gentner, T. Jost, W. Wang, S. Zhang, A. Dammann, and U.-C. Fiebig, "Multipath assisted positioning with simultaneous localization and mapping," *IEEE Transactions on Wireless Communications*, vol. 15, no. 9, pp. 6104–6117, 2016.

[18] M. Ulmschneider, C. Gentner *et al.*, "Multipath assisted positioning for pedestrians using LTE signals," *Mobile Information Systems*, vol. 2017, p. 9170746, 2017. [Online]. Available: https://www.hindawi.com/journals/misy/2017/9170746/

[19] E. Leitinger, L. Wielandner, A. Venus, and K. Witrisal, "Multipath-based SLAM with cooperation and map fusion in MIMO systems," *arXiv preprint arXiv:2405.02126*, 2024. [Online]. Available: https://arxiv.org/abs/2405.02126

[20] X. Li, X. Cai, E. Leitinger, and F. Tufvesson, "A belief propagation algorithm for multipath-based SLAM with multiple map features: A mmWave MIMO application," *arXiv preprint arXiv:2403.10095*, 2024. [Online]. Available: https://arxiv.org/abs/2403.10095

[21] Y. Xiong, F. Liu, Y. Cui, W. Yuan, T. X. Han, and G. Caire, "On the fundamental tradeoff of integrated sensing and communications under Gaussian channels," *IEEE Transactions on Information Theory*, vol. 69, no. 9, pp. 5723–5751, 2023.

[22] Han, J. Choi, and R. W. Heath, "Radar imaging based on IEEE 802.11ad waveform in V2I communications," arXiv preprint arXiv:2208.02473, 2022. [Online]. Available: https://arxiv.org/abs/2208.02473